Whole-Body Model Predictive Control for Industrial Humanoids

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31 March 2026

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Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

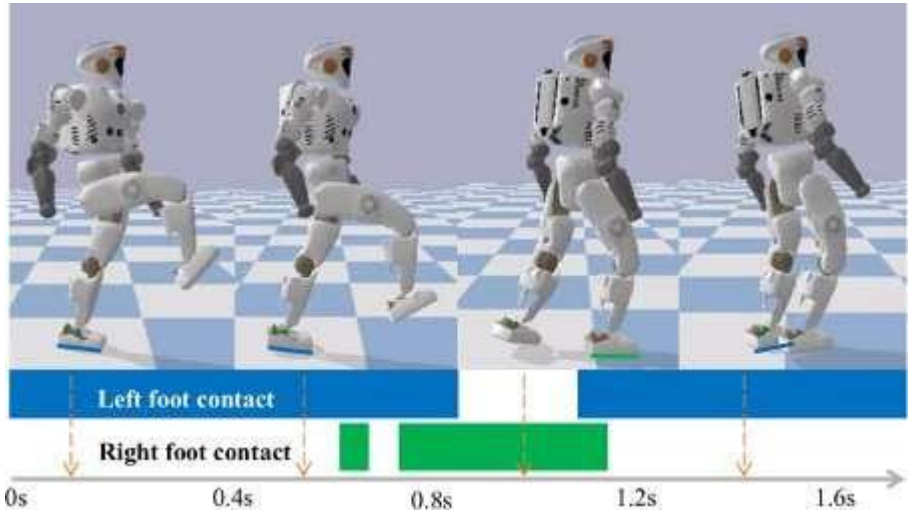
We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.



Whole-Body Model Predictive Control for Industrial Humanoids

Overview

- This project focuses on developing a control system for a humanoid robot that enables stable movement and task execution using Model Predictive Control (MPC).
- Humanoid robots are designed to operate in human environments, but maintaining balance while performing tasks is a major challenge. This project explores how predictive control techniques can be used to plan and optimize robot motion in real time.
- Since this is a bachelor-level project, the work will be conducted in simulation, with a simplified representation of humanoid dynamics.



Prof. Boris, Prof. Prem and Muhammad Saeed

Whole-Body Model Predictive Control for Industrial Humanoids

Project Objective

The objective is to design a predictive control system that allows a humanoid robot (or simplified model) to:

- Maintain balance
- Move toward a target
- Carry out simple tasks (e.g., move or carry an object)
- Respond to disturbances

Instead of controlling a full humanoid from the beginning, the project will focus on simplified dynamic models, such as:

- inverted pendulum model
- center-of-mass control
- simplified walking or balancing system

Whole-Body Model Predictive Control for Industrial Humanoids

Key Research Areas

- **Model Predictive Control (MPC)**
  - Optimization-based control
  - Predicting future system behavior
  - Receding horizon control
- **Robot Dynamics**
  - Basic kinematics and dynamics
  - Center of mass (CoM)
  - Stability concepts
- **Humanoid Balance**
  - Zero Moment Point (ZMP)
  - Inverted pendulum models
- **Simulation-Based Control**
  - Control implementation in simulation environments

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System Architecture

The system follows a **closed-loop predictive control pipeline**:

State Estimation

- Current system state (position, velocity, angle)
- Example: pendulum angle, cart position

Prediction Model

- Mathematical model predicts future states
- Uses system dynamics

MPC Optimizer

- Solves optimization problem over a time horizon
- Minimizes deviation from desired state

Control Input

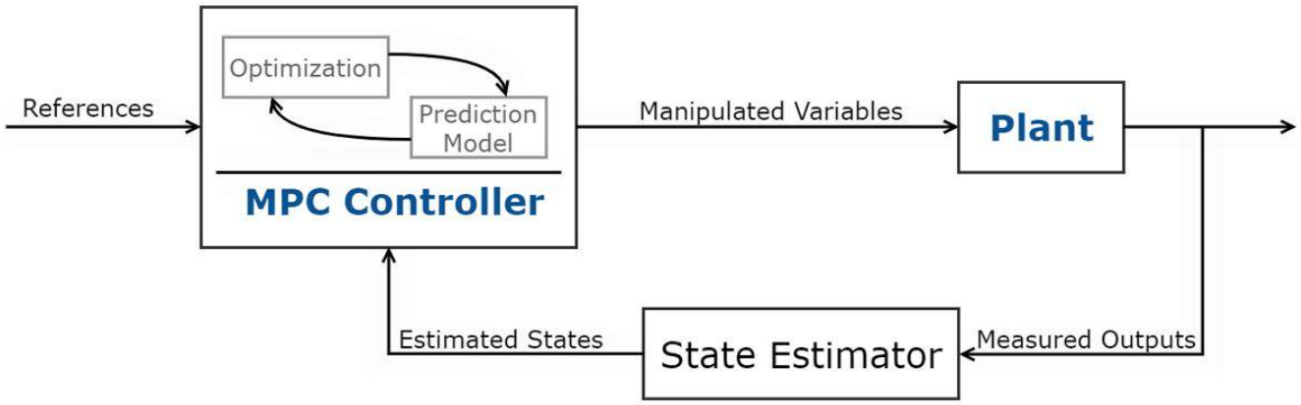
Computes optimal control action (force/torque)

Simulation Environment

- Applies control to system
- Updates system state

Feedback Loop

- New state is fed back into controller



Key Idea:

Predict → Optimize → Act → Repeat

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Research Plan and Strategy

Structured Approach (Bachelor-Friendly)

- Start with basic control concepts, then move to predictive control
- Focus on understanding behavior, not just implementation
- Use simplified dynamic models instead of full humanoids

Development Strategy

Foundation

- Study MPC basics and inverted pendulum model
- Understand stability and control objectives

Baseline System

- Implement simple controller (PID or rule-based)
- Use it as comparison reference

MPC Development

- Build prediction model
- Define cost function and constraints

Integration

- Combine model + optimizer + simulation

Testing & Analysis

- Evaluate stability, robustness, efficiency

Strategy Principle:

Start simple → Validate → Improve → Compare

Simulation Model

The project will be based on a **simplified simulation model** representing an industrial workspace.

The simulation environment will include:

- A defined workspace (2D or simple 3D)
- A robot agent capable of performing basic actions
- Multiple objects such as colored blocks or components
- Target locations such as trays or workstations

The robot's task is to interpret instructions and interact with the environment accordingly.
The simulation will include:

- object manipulation tasks (pick-and-place)
- spatial reasoning (left, right, near)
- task sequencing

Tools and Technologies

Category	Tools	Purpose
Programming	Python	Main development
Numerical Computing	NumPy / SciPy	Mathematical modeling
Optimization	CVXPY	MPC optimization solver
Simulation	PyBullet / custom simulation	System simulation
Visualization	Matplotlib	Plot results
Development	Jupyter / VS Code	Experimentation

Notes:

- Start with custom Python simulation
- Use PyBullet only if time allows
- Focus on understanding MPC, not tools

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Evaluation Strategy

Metric	Description	Purpose
Stability	Ability to keep system balanced	Core performance
Tracking Error	Distance from target position	Accuracy
Response Time	Time to stabilize system	Efficiency
Overshoot	Excess deviation from target	Control quality
Robustness	Performance under disturbance	Reliability
Control Effort	Amount of force used	Energy efficiency
Comparison	Baseline vs MPC	Demonstrate improvement

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Expected Deliverables

Deliverable	**Description**	**Outcome**
Literature Review	Study MPC and control basics	Concept understanding
Simulation Model	Inverted pendulum system	Test platform
Baseline Controller	PID / simple controller	Reference system
MPC Controller	Predictive control implementation	Core contribution
Experimental Results	Performance evaluation	Analysis
Visualizations	Graphs of system behavior	Interpretation
Final Report	Methodology + results	Academic output
Demonstration	Simulation showing balance control	Project showcase

Whole-Body Model Predictive Control for Industrial Humanoids

Literature

- Model Predictive Control: Theory, Computation, and Design
- Biped Walking Pattern Generation by using Preview
- https://github.com/fan-ziqi/rl_sar
- https://github.com/NVlabs/SOMA-X
- https://research.nvidia.com/labs/sil/projects/kimodo/
- https://github.com/NVlabs/GEM-X
- https://seed-viewer.bones.studio/
- https://github.com/NVIDIA/soma-retargeter/
- https://github.com/NVlabs/ProtoMotions
- https://nvlabs.github.io/GEAR-SONIC/

you can start exploring this topic and building the simulation frameworks.
https://bones.studio/datasets/seed
Please use this as a benchmark for setting up simulation frameworks.
https://huggingface.co/nvidia/GEAR-SONIC



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